Meteorite Landings: A Data Visualisation Analysis

by

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Table of Content

Declaration of Authorship..... 2

Table of Content..... 3

Table of Figures.....4

Introduction.....5

Dataset..... 6

Fig. 1.2: Missing Values in the Longitude column (reclong)..... 9

Key Findings and Visualisations..... 10

Fig. 1.3: Geospatial Bubble Map of Meteorite Landings in Shiny..... 11

Fig. 1.4: Geospatial Bubble Map of Meteorite Landings for the United States of America..... 11

Fig. 1.5: Distribution of Discovery Type of Meteorites..... 12

Fig. 1.6: Distribution of Meteorite Mass by Discovery Type..... 13

Fig. 1.7: Discoveries per year, Fell vs Found.....14

Fig. 1.8: Top 10 Meteorite Classes Found Since 2000.....15

Design Considerations and Complex Implementations..... 15

Conclusion..... 16

References..... 18

Appendix..... 19

Table of Figures

Fig. 1.2: Missing Values in the Longitude column (reclong)..... 9

Key Findings and Visualisations..... 10

Fig. 1.3: Geospatial Bubble Map of Meteorite Landings in Shiny..... 11

Fig. 1.4: Geospatial Bubble Map of Meteorite Landings for the United States of America..... 11

Fig. 1.5: Distribution of Discovery Type of Meteorites..... 12

Fig. 1.6: Distribution of Meteorite Mass by Discovery Type..... 13

Fig. 1.7: Discoveries per year, Fell vs Found..... 14

Fig. 1.8: Top 10 Meteorite Classes Found Since 2000..... 15

Introduction

This report aims to investigate the patterns in meteorite landings across Earth. The dataset was obtained from the NASA Open Data Portal, with 45,716 records. The analyses were conducted using R with packages such as dplyr, tidyverse, and psych, and a shiny R app was created for interactivity and to showcase visualisations such as bubble maps, box plots, density plots, and line plots. e2eSA (2023) published a visualisation on this NASA dataset to answer a couple of research questions, particularly towards areas that got hit by meteorites and their discovery types. The technology used for this analysis was Tableau **(question b)**. Geospatial maps were created, and the United States was also investigated to prove that the meteorites were found more towards more populated areas. In their map, the colours of the points were identified as the classes of each of the meteorites. This analysis builds on that work by using an interactive Shiny app, rather than Tableau.

Edward Tufte is a statistician and professor emeritus at Yale University who is widely regarded as a pioneer in data visualisation ("Edward Tufte," 2025). In 2009, he outlined five core principles for presenting information effectively. These principles include showing comparison, answering questions (showing causality), displaying multiple variables, providing a plurality of evidence, and documenting sources for credibility. The visualisations implemented in this analysis were created with Tufte's principles in mind.

The points on the geospatial map represent the discovery type of each of the meteorites, rather than their classes. Also, a column plot was created to find the meteorite class that was found the most since the year 2000. A density plot was also implemented to show a comparison between the mass of meteorites that were found versus those that were witnessed. This study also recognises patterns in the missingness of the dataset, specifically the geolocation columns (reclat and reclong). The code implementation was guided by documentation such as r-blogger, ropensci, ggplot2, r-spatial, and datacamp R documentation for geospatial maps and the sf package documentation for spatial filtering approach **(question a)**. It also provides visual proof that the missingness of these variables

(reclat and reclong) is Missing At Random (MAR), as they are missing in recent years **(question c).**

Dataset

The dataset consists of 10 columns, including meteorite name, meteorite IDs, classification type (recclass), size of meteorites (mass), discovery type (fall), year of discovery, and geometric coordinates (reclat, reclong and geolocation). For the nametype column, the relict category represents meteorites that were once meteorites but are now altered by atmospheric pressure on the Earth, while the valid category represents meteorites. There are a total of 15,052 missing values, with mass (grams) having 131 missing values, year having 291 missing values, and reclat and reclong having 7,315 missing values. In the dataset, a pattern was recognised in the missingness of the geospatial coordinate columns (reclat and reclong), which is at random (Missing At Random, MAR). This missing data is more towards recent years, particularly between 1981 and 2010 **(question f)**.

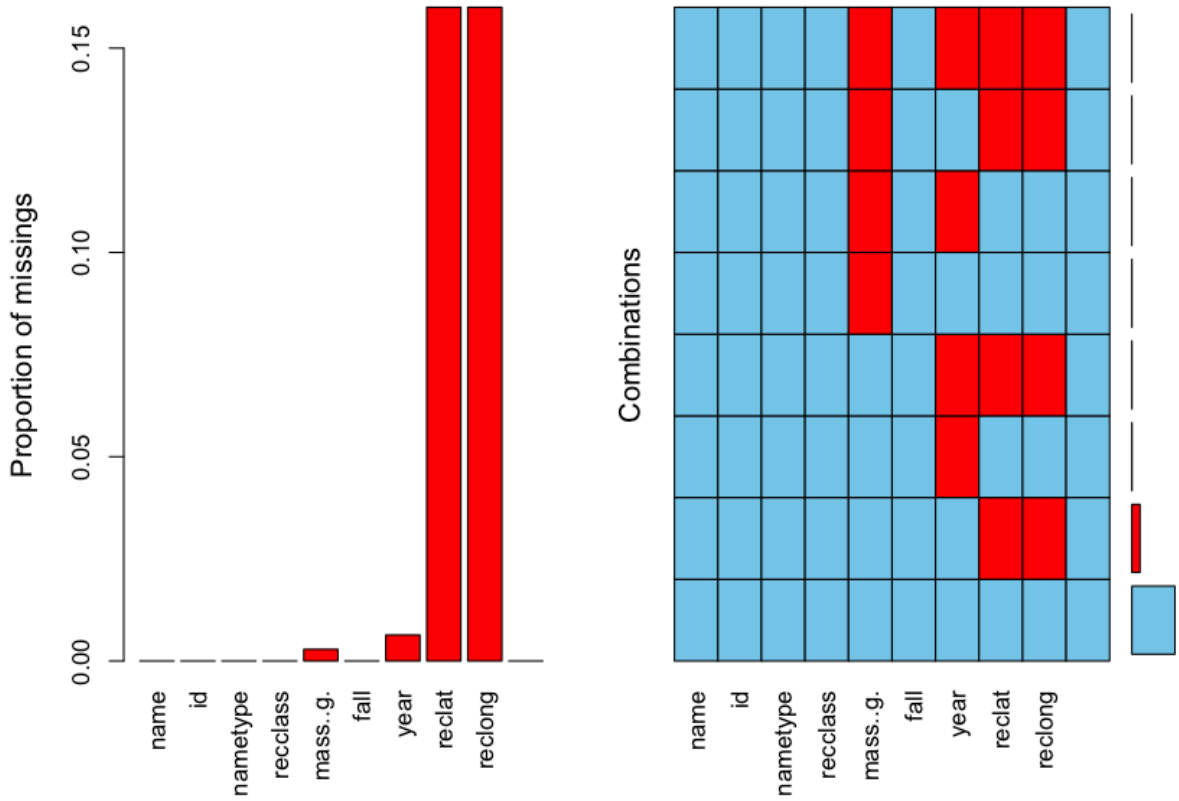

Fig. 1.0: Proportion of Missingness in the dataset

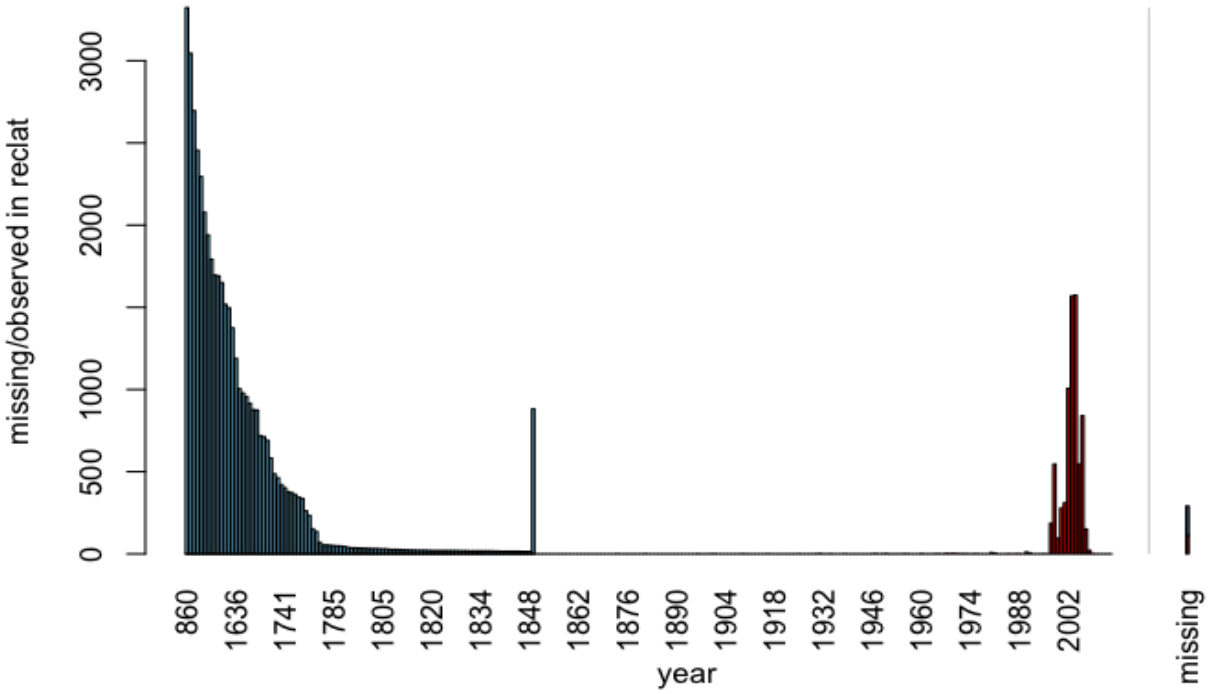

Fig. 1.1: Missing Values in the Latitude column (reclat)

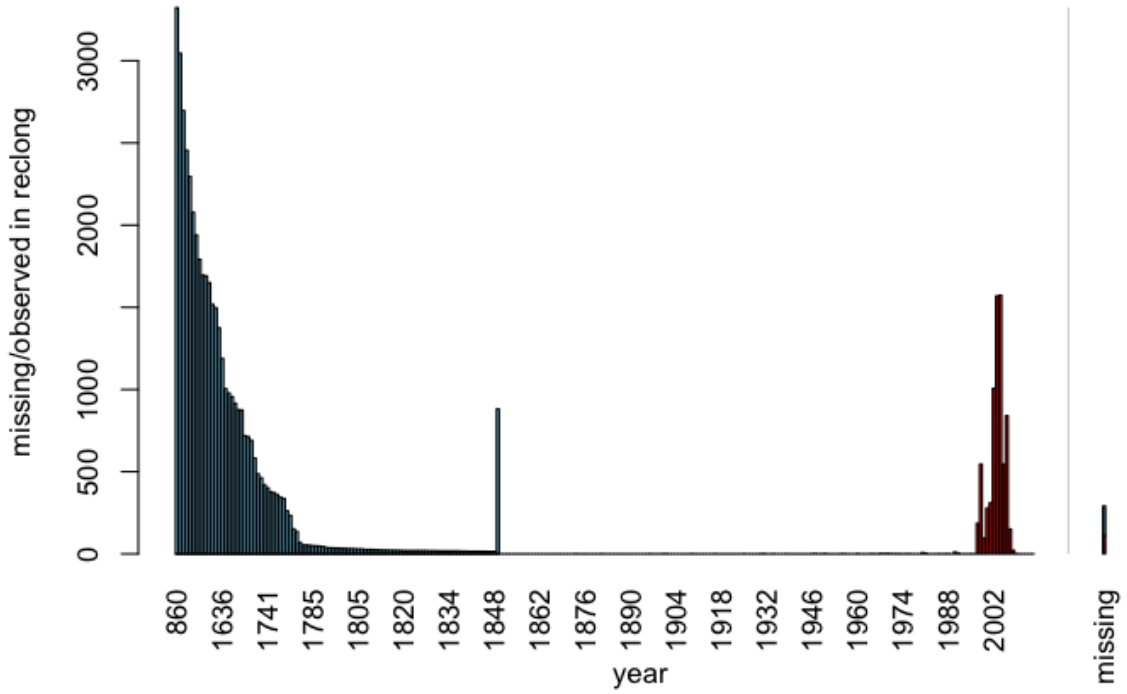

Fig. 1.2: Missing Values in the Longitude column (reclong)

Based on this pattern, a decision had to be made on dealing with the missing values. These values were not removed because various other visual plots will require specific columns, and removing these data could also mean removing important data for these visualisations. Removing these observations may lead to bias in the data and could distort other visualisations made.

The mass (in grams) column was also investigated. This column was heavily positively skewed, in which the larger values may have more effect on plotting (geospatial bubble map plot). In this dataset, there are more found meteorites than witnessed, with 44,609 found and 1,107 witnessed.

Key Findings and Visualisations

A geospatial bubble map was created to show the locations where each meteorite was discovered and how it was discovered. The salmon bubble indicates the meteorite was witnessed, while the teal bubble indicates it was found. For interactivity, a radio button was created in Shiny to select a view of the bubble map for only meteorites that were witnessed, found or a combination of both. A year range slider component was also created to filter the year limit to display on the map. It will give a view of what meteorites were found or witnessed in the years selected.

The map displays a pattern in these meteorite landings. It indicates that the meteorites were found in densely populated areas. This suggests that there may be more eyewitnesses or more interest in meteor research, as well as facilities to explore the meteors. This visualisation choice aligns with Edward Tufte's principle of the importance of displaying multiple variables in a single plot. The map showcases the discovery type of each meteorite (fall), the size of these meteorites (mass) represented as the size of the bubbles, and the geolocation where they landed (reclat and reclong).

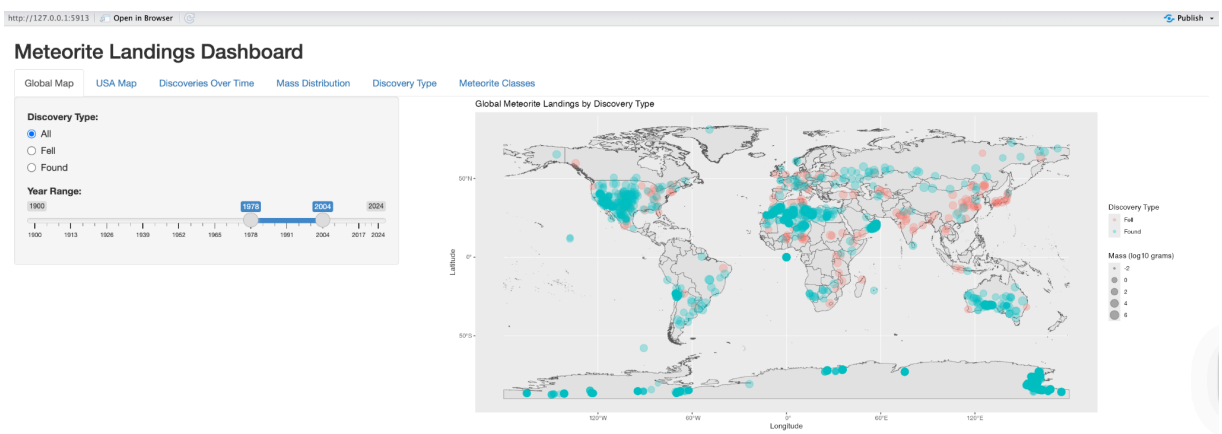

Fig. 1.3: Geospatial Bubble Map of Meteorite Landings in Shiny

The meteorite landings needed further investigation, particularly in densely populated areas, and so a geospatial bubble map of the United States of America was created to give a broader view of where these meteorites landed.

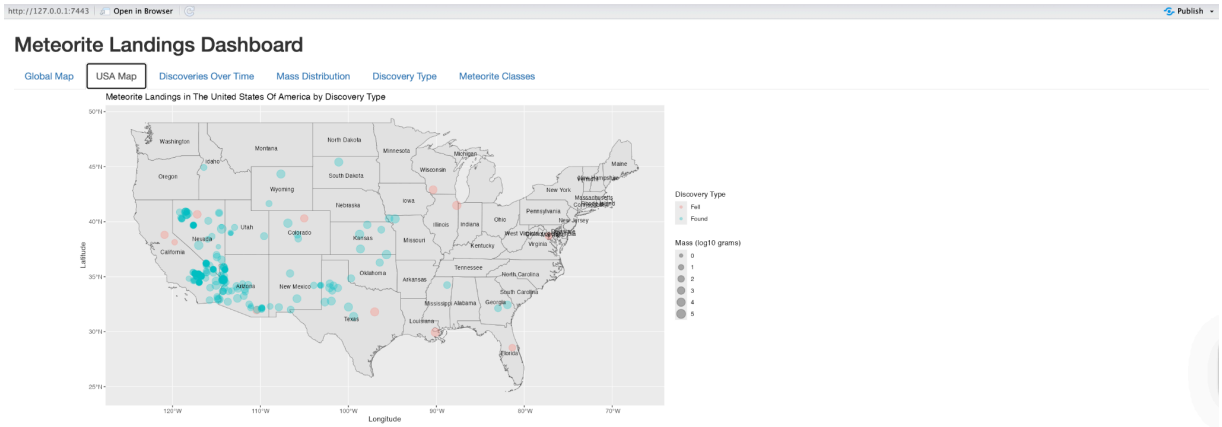

Fig. 1.4: Geospatial Bubble Map of Meteorite Landings for the United States of America

Edward Tufte's second principle states that visualisation should show causality (Edward Tufte, 2009, 31:00). This means plots should show why something happened and not just show what happened. The USA map shows that the meteors landed more in California and Arizona. This may be due to more eyewitnesses, as California is one of the three largest states in the US, and Arizona has been one of the fastest-growing states since 2000. Moreover, the map shows that the majority of meteorites that landed in the US were found rather than witnessed. This is consistent with the landing pattern identified in the geospatial map. The

map was created using the sf spatial filtering method, and it ensured that only meteorites within the US border polygon were displayed.

The distribution of the discovery type of the meteorites was considered. The bar plot shows a comparison of the number of meteorites classified as found against the number of meteorites witnessed, with the meteorites found being 97.6% of the meteorites recorded (44,609) and the meteorites witnessed being 2.4% (1,107). This supports the landings on the geospatial bubble map, which shows that meteorites are not witnessed more frequently in populated areas because they fall there, but because there are more eyewitnesses present to find them after they fall. This visualisation adheres to Edward Tufte's first principle of comparison (Tufte, 2009, 27:00). The bar chart was chosen because the discovery type column is categorical with only two values, and therefore, a simple visualisation that displays count comparison was the most appropriate **(question e)**.

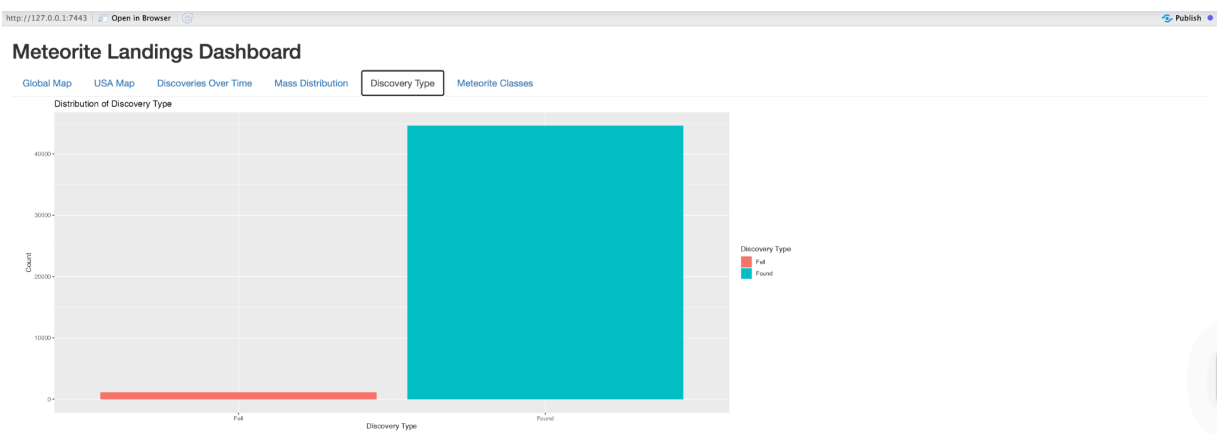

Fig. 1.5: Distribution of Discovery Type of Meteorites

The distribution of each meteorite mass by discovery type was also investigated. This plot indicates that the witnessed meteorites are heavier than the meteorites found. This is because larger meteorites are harder to miss compared to those found on the ground. Goodall (2023) said that large meteorites cause loud boom noises, such as a thunder noise, and for this reason, meteorites can be more easily witnessed than found.

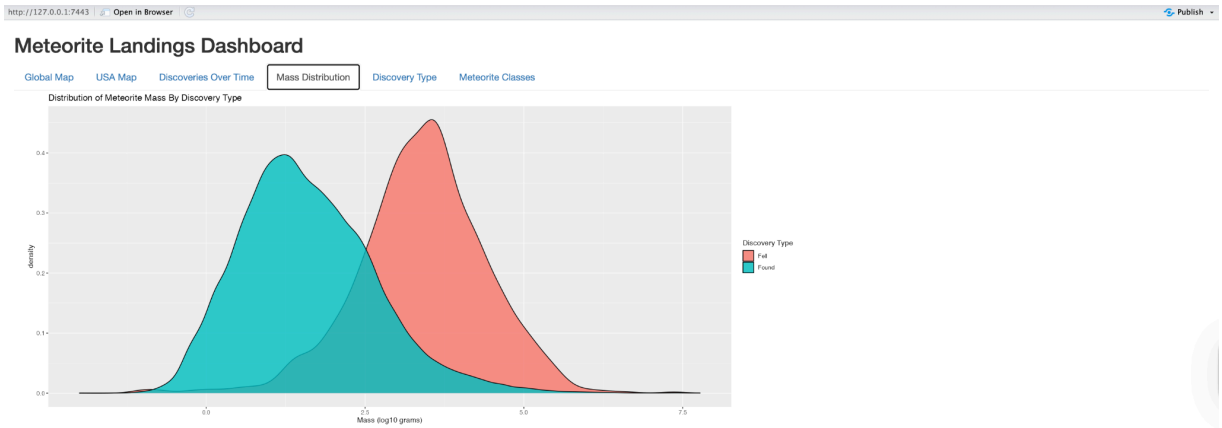

Fig. 1.6: Distribution of Meteorite Mass by Discovery Type

According to Tufte's first principle, a good visualisation should show comparison (Tufte, 2009, 27:00). The density plot achieves this by placing the mass distributions of witnessed versus found meteorites on the same axis. The mass (in grams) column was extremely positively skewed (skewness = 76.91), and for this reason, a log10 transformation was made to reduce the larger values into a smaller range of values before plotting **(questions e, f)**.

The discoveries per year by discovery type were also examined using a line plot. The trend of both categories (found and witness) was relatively low until around the 1970s, when the found category had a sharp uptrend. It peaked in the early 2000s before declining. The

witnessed category stays roughly at the same level throughout the years. The sharp increase in the early 80s was due to the Antarctic Search for Meteorites (ANSMET) programme funded from 1976 to 1977, where Russian, Japanese and American geologists have found up to 20,000 meteorites since that time ("ANSMET," 2025). The witnessed category remains roughly the same throughout the years, while organised searches have increased the number found afterwards. This visualisation adheres to Tufte's second principle, which is to show causality (Edward Tufte, 2009, 31:00). The line chart explains why the meteorite discovery numbers change over time rather than showing that they changed **(question f)**. The referencing of the ANSMET programme also answers to Tufte's fifth principle, which emphasises the importance of showing credibility and providing documentation and explanations to support claims made from a dataset (Edward Tufte, 2009, 37:30).

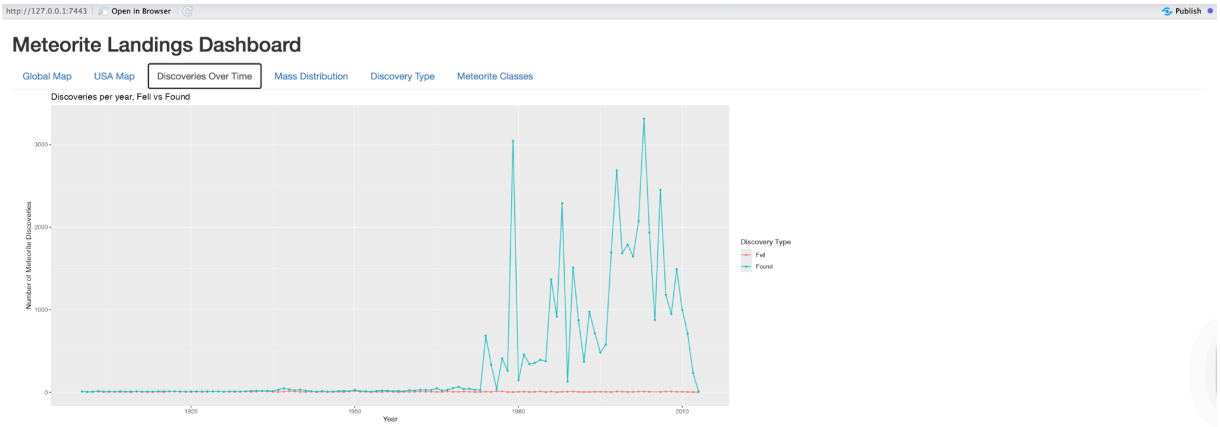

Fig. 1.7: Discoveries per year, Fell vs Found

The visualisation below depicts the top 10 meteorite classes observed since 2000. Amongst all the meteorite classes, L6 was the most recorded at 8,240 across all the years. It was observed that 2003 produced the highest number of meteorites found or witnessed, with the

L5 meteorite class the most frequently recorded at that time. However, there was a sharp decline in the number of meteorites recorded between 2004 and 2005, as shown in the line chart. This visualisation adheres to Tufte's third principle of showing multiple variables in a plot (Edward Tufte, 2009, 33:00). It displays multiple variables simultaneously, which are the year when meteorites were found or witnessed, the meteorite class, and the number of discoveries per year **(question e)**.

Fig. 1.8: Top 10 Meteorite Classes Found Since 2000

Design Considerations and Complex Implementations

The complexities encountered in this analysis are mainly in data filtering, decisions made towards handling missing data, and the implementation of the Shiny app. The most complex implementation in this analysis was filtering the sf spatial data to clip geolocation points to the United States border. The dataset was converted from a dataframe to an sf spatial object using the **st_as_sf** function, which combines the reclat and reclong columns into a single geometry column for spatial operations. The geometry points were compared with the

polygon shape of the US border using the **st_filter** function. This was used to keep only the points that fall within the US map **(question d)**. The shiny app also had its complications, such as selecting the best component for the geospatial map interactions based on filtered data. A reactive block was used to filter the data based on user inputs. A radio button was created for the discovery type to select either the witness category or the found category. A slider component was created to allow the users to select a year range for the map visualisation so that it updates dynamically whenever the user changes their selection **(question d)**.

The design considerations in this analysis include missing data handling, log10 transformation, and Alaska and Hawaii filtering. There was difficulty with making the best choice for missing values, but the final verdict was to only filter those datasets when implementing visualisations. This is because different visualisations require different variables. If these records are removed permanently, they may introduce bias to the dataset and cause important patterns to be lost. The log10 transformation was applied to the mass column because it has extremely positive skewness (skewness = 76.91). Without this transformation, the few extremely large meteorites would produce enormous bubbles, making all other points invisible on the map **(question e)**. The US map was also filtered to provide a larger view of the map. This is because Alaska and Hawaii are geographically distant from the US continent. For this reason, the size of the map was zoomed out so far that the 48 states became too small to view clearly, and as a result, Alaska and Hawaii were filtered out from the US map **(question e)**.

Conclusion

In conclusion, the visualisations show that meteorites were found in more populated areas such as the United States and India, indicating that there are more eyewitnesses and more facilities and organisations to explore these meteorites. Also, it was observed that the meteorites witnessed are heavier compared to those that were found. The sharp increase in the number of meteorites found on the line plot was due to the development of the ANSMET programme from 1976, and it has recovered over 20,000 meteorites from Antarctica since

that time. The Shiny app adds interactivity through a radio button component to filter the discovery type of the meteorites and a year range slider to filter the years they were recorded on the global map. If more time were given, the Shiny app and visualisations could be more interactive. A small window component could be added to the geospatial map when users hover over the points to provide details such as the year recorded, size of meteorites, discovery type, and so on **(question g)**. There should be more emphasis on interactivity over all other visualisations aside from the map plots. Outliers in the dataset could also be investigated, particularly in the year column. The data error of the year 2101 identified during analysis could be investigated to determine the source **(question g)**. This analysis demonstrates that data visualisation can reveal patterns in meteorite landings by uncovering the human factors behind the dataset.

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Save a ggplot (or other grid object) with sensible defaults — ggsave • ggplot2 [WWW Document], n.d. URL https://ggplot2.tidyverse.org/reference/ggsave.html (accessed 3.6.26).

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Appendix

```
#
# This is a Shiny web application. You can run the application by clicking
# the 'Run App' button above.
#
# Find out more about building applications with Shiny here:
#
#    https://shiny.posit.co/
#

library(shiny)
library(psych)
library(ggplot2)
library(dplyr)
library(VIM)
library(tidyverse)

# Did not work
#install.packages("summarytools", dependencies = TRUE)
#library(summarytools)



# Reference:
https://www.r-bloggers.com/2024/10/top-10-r-packages-for-exploratory-data-analysis-eda-b
ookmark-this/
install.packages("skimr")
library(skimr)
```

```
library(leaflet)
library(sf)

install.packages(c("rnaturalearth", "rnaturalearthdata"))
library(rnaturalearth)
library(rnaturalearthdata)



# Reference: https://docs.ropensci.org/rnaturalearthhires/

install.packages(
  "rnaturalearthhires",
  repos = "https://ropensci.r-universe.dev",
  type = "source"
)

library(rnaturalearthhires)





df <- read.csv("Meteorite Landings.csv")

# 45,716 rows, 10 columns
#dim(df)
skim(df)
```

```
View(df)


# Strong skewness on mass, reclat, and reclong variables.
# Valid meteor name is 99.8% - 45641 and relict names are .2% - 75 of them
summary(df)
describe(df)


# Converting all character columns to factors. lapply returns a list, whihc is good for a
# dataframe

df <- data.frame(lapply(df, function(x) {
  if (is.character(x)) as.factor(x)
  else x
}))

# Columns mass.g., year, reclat, reclong have missing values.
# reclat and reclong has 7315 missing values, year has 291 missing values,
# and mass has 131 missing values

colSums(is.na(df))


# Total missing values - 15,052
sum(is.na(df))
```

```
# There are missing values in mass, latitude, longitude variables
a <- aggr(df)
a

# save proportion of missingness plot
quartz.save("proportion_of_missingness.png")


par(mfrow=c(1,1))

# Lat
yr_v_lat <- df[, c("year","reclat")]

# Missing latitude values from recent years (1981-2010) - MAR
barMiss(yr_v_lat)
quartz.save("missing_data_in_reclat.png")

marginplot(yr_v_lat)

# Long
yr_v_lon <- df[, c("year","reclong")]

# Missing longitude values from recent years (1997-2010)
barMiss(yr_v_lon)

# save barMiss plot
quartz.save("missing_data_in_reclong.png")

marginplot(yr_v_lon)
```

```
# For handling dataset, We
# omit them during visualisation because it is does not make sense
# to use imputation on latitude and longitude dataset. We do not know the exact
# co-ordinates where they landed


df %>%
  filter(is.na(reclat)) %>%
  select(GeoLocation) %>%
  head(10) %>%
  pull(GeoLocation)

table(df$nametype)
table(df$fall)

# World and USA maps loaded once at startup (not inside server)
world <- ne_countries(scale = "large", returnclass = "sf")
usa <- ne_states(returnclass = "sf", country = "United States of America")

# Filtering out Alaska and Hawaii because they are far away from most states of the usa and it
will make
# the map small

usa_filter <- usa %>% filter(!name %in% c("Alaska", "Hawaii"))
```

```
# Top 10 meteorite classes
recent_met <- df %>%
  filter(!is.na(year)) %>%
  count(recclass) %>%
  arrange(desc(n)) %>%
  head(10)


# UI block

ui <- fluidPage(
  titlePanel("Meteorite Landings Dashboard"),

  tabsetPanel(

    # Tab 1: Global Map
    tabPanel("Global Map",
             sidebarLayout(
               sidebarPanel(
                 radioButtons("fall_filter", "Discovery Type:",
                              choices = c("All", "Fell", "Found"),
                              selected = "All"),
                 sliderInput("year_range", "Year Range:",
                             min = 1900, max = 2024,
                             value = c(1900, 2024), sep = "")
               ),
               mainPanel(plotOutput("global_map", height = "500px"))
             )
```

```
    ),

    # Tab 2: USA Map
    tabPanel("USA Map",
             mainPanel(plotOutput("usa_map", height = "500px"))
    ),

    # Tab 3: Discoveries Over Time
    tabPanel("Discoveries Over Time",
             mainPanel(plotOutput("line_chart", height = "500px"))
    ),

    # Tab 4: Mass Distribution
    tabPanel("Mass Distribution",
             mainPanel(plotOutput("density_plot", height = "500px"))
    ),

    # Tab 5: Discovery Type
    tabPanel("Discovery Type",
             mainPanel(plotOutput("bar_chart", height = "500px"))
    ),

    # Tab 6: Meteorite Classes
    tabPanel("Meteorite Classes",
             mainPanel(plotOutput("col_chart", height = "500px"))
    )

  )
)
```

```
# Server Block
server <- function(input, output) {

  # Reactive filtered data for global map
  filtered_data <- reactive({
    data <- df %>%
      filter(!is.na(reclat) & !is.na(reclong),
             year >= input$year_range[1],
             year <= input$year_range[2])

    if (input$fall_filter != "All") {
      data <- data %>% filter(fall == input$fall_filter)
    }
    data
  })

  # 1.) Meteorites found since 2000

  output$global_map <- renderPlot({
    ggplot(data = world) +
      geom_sf() +
      geom_point(data = filtered_data(),
                 aes(x = reclong, y = reclat,
                     size = log10(mass..g.),
                     color = fall),
                 alpha = 0.3) +
      labs(
```

```
      x = "Longitude",
      y = "Latitude",
      title = "Global Meteorite Landings by Discovery Type",
      color = "Discovery Type",
      size = "Mass (log10 grams)"
    )
  })

  # 2.) Distribution of discovery type
  output$usa_map <- renderPlot({
    po <- df %>%
      filter(!is.na(reclat) & !is.na(reclong),
             mass..g. > 0, year >= 2000) %>%
      st_as_sf(coords = c("reclong", "reclat"), crs = st_crs(4326))

    usa_points <- st_filter(po, usa_filter)

    # Add text to map:
https://r-spatial.org/r/2018/10/25/ggplot2-sf-2.html#:~:text=States%20(polygon%20data),w
here%20to%20draw%20their%20names.

    ggplot(data = usa_filter) +
      geom_sf() +
      geom_sf(data = usa_points,
              aes(size = log10(mass..g.), color = fall),
              alpha = 0.3) +
      geom_sf_text(data = usa_filter, aes(label = name), size = 3) +
      labs(
        x = "Longitude",
```

```
    y = "Latitude",
    title = "Meteorite Landings in The United States Of America by Discovery Type",
    color = "Discovery Type",
    size = "Mass (log10 grams)"
  )
})

# 3.) Bubble Map showing where meteors fell



# Reference: https://r-spatial.org/r/2018/10/25/ggplot2-sf.html
output$line_chart <- renderPlot({
 df %>%
  group_by(year) %>%
  filter(year >= 1900 & year <= 2024) %>%
  count(fall) %>%
  ggplot(aes(x = year, y = n, color = fall)) +
  geom_line() +
  geom_point(size = 0.8) +
  labs(
    x = "Year",
    y = "Number of Meteorite Discoveries",
    title = "Discoveries per year, Fell vs Found",
    color = "Discovery Type"
  )
})

# 4.) Specific Bubble Map (USA) - Where
```

```
# Reference: https://www.rdocumentation.org/packages/sf/versions/1.0-23/topics/st_as_sf
output$density_plot <- renderPlot({
  df %>%
    filter(mass..g. > 0, !is.na(mass..g.)) %>%
    ggplot(aes(x = log10(mass..g.), fill = fall)) +
    geom_density(alpha = 0.8) +
    labs(
      x = "Mass (log10 grams)",
      title = "Distribution of Meteorite Mass By Discovery Type",
      fill = "Discovery Type"
    )
})

# 5.) Density Plot - Distribution of meteorite mass by discovert type
output$bar_chart <- renderPlot({
  df %>%
    ggplot(aes(x = fall, fill = fall)) +
    geom_bar() +
    labs(
      x = "Discovery Type",
      y = "Count",
      title = "Distribution of Discovery Type",
      fill = "Discovery Type"
    )
})

# 6.) Line plot - Discoveries per year as a line, split by Fell vs Found.
```

```
  # Fell: Witnessed meteorites have roughly been the same over time.
  # Found: Fluctuates over time. Spiked around 1980 and dropped after 2010.

  output$col_chart <- renderPlot({
    df %>%
      filter(year > 2000, !is.na(year), recclass %in% recent_met$recclass) %>%
      count(year, recclass) %>%
      ggplot(aes(x = year, y = n, fill = recclass)) +
      labs(
        title = "Top 10 Meteorite Classes Found Since 2000",
        x = "Year",
        y = "Number of Meteorites"
      ) +
      geom_col()
  })

}


shinyApp(ui, server)
```